Lecture 03

Procedural Abstraction and Testing

EECS 280

Prof. John Kloosterman

Announcements

Are you a student in **EECS 183, 203, 280, or 281** interested in **tutoring** or **mentorship?**

Become a RENEW CS mentee!

RenewCS is a **FREE** student-led mentorship and tutoring program for students in Computing. Students can expect individual and group mentorship with junior or senior CSE students, in-person and remote tutoring sessions, and community-building events.

Apply for the program here:
Mentorship applications close **1/23!**

myumi.ch/dg1XV

Class outline

- Saying hi

- Diagnostic quiz
- Mini-lecture
- Exercises on laptops

Meeting each other

- 2 rounds of finding someone you don't know
  - Say hi
  - What was your path into EECS 280 at Michigan?

Diagnostic

Individual Time

Compare with partner

- Move if you need to

Question 1

```
void fn(int x) {
   if (x > 0) {
      cout << x << endl;
      fn(x - 1);
   }
}

int main() {
   fn(2);
}
```

Question 2

```
// EFFECTS: returns true if and only if the numbers in v are in
//   increasing order.
bool is_increasing(vector<int> v) {
   for (int i = 0; i < v.size() - 2; i++) {
      if (v.at(i) >= v.at(i + 2)) {
         return false;
      }
   }
   return true;
}
```

Coding exercise

Coding Exercise

- Download and set up following instructions at jkloosterman.net/280_transfer

- The autograder will test your code and your tests
  - 10 points for tests
  - 5 points for your implementations
- Submit to autograder even before done!

- Equations are on your diagnostic sheet

During exercise time

- Verify computer setup with VS Code
- Verify AI turned off

Homework

- Submit exercise to autograder by tonight 11:59PM
- **There is** a textbook reading due Sunday 11:59PM

- P1: on track if you have implementations and tests for the simple functions in stats.cpp